

Disproof of conjecture 00000001075

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Verdict: FALSE

We disprove conjecture 00000001075 as stated. Over the field $\mathbb{F}_5$ the function

$$f = \delta_0 - \delta_1 = (1, -1, 0, 0, 0)$$

satisfies $\#\text{supp}(f) + \#\text{supp}(\widehat{f}) = 6 = q + 1$, so it is an equality configuration; but its support pair is $(2, 4)$, not the claimed $(q, 1) = (5, 1)$, and f is not a Fourier translate of any affine function $a \cdot x + b$. An exhaustive enumeration of all $3^5 = 243$ functions $\mathbb{F}_5 \rightarrow \{0, \pm 1\}$ shows that there are exactly 32 equality configurations, 20 of which have support pair $(2, 4)$ and none of which is an affine Fourier translate. Hence both the parenthetical “(support pairs of q and 1)” and the clause “no other equality configurations exist” fail.

1 The conjecture

Definition 1 (entropic uncertainty for the finite-field DFT). For a prime power q and $f : \mathbb{F}_q \rightarrow \mathbb{C}$ write

$$\widehat{f}(\xi) = \sum_{x \in \mathbb{F}_q} f(x) \omega^{\xi x}, \quad \omega = e^{2\pi i/q},$$

for its finite-field Fourier transform, and let $\text{supp}(f)$ denote the set of points at which f is nonzero. The entropic uncertainty principle asserts

$$\#\text{supp}(f) + \#\text{supp}(\widehat{f}) \geq q + 1. \quad (1)$$

The conjecture under test adds an exact classification of the equality cases.

Quoted verbatim from `conjectures/00000001075.md`:

English. Definition: The entropic uncertainty principle for the finite-field Fourier transform states $\#\text{supp}(f) + \#\text{supp}(\widehat{f}) \geq q + 1$. Conjecture: Equality holds exactly for Fourier translates of affine functions $a \cdot x + b$ (support pairs of q and 1), and no other equality configurations exist.

$$\begin{array}{ccc} \widehat{f} & & \#\text{supp}(f) + \#\text{supp}(\widehat{f}) \geq q + 1 \\ a \cdot x + b & \text{Fourier} & q \quad 1 \end{array}$$

The conjecture therefore makes two assertions: (i) every equality configuration is a Fourier translate of an affine function and has support pair $(q, 1)$ (equivalently $(1, q)$), and (ii) there are no other equality configurations. Both are refuted below, over the prime $q = 5$.

2 Exact arithmetic in $\mathbb{Z}[\omega]$

Throughout we take $q = 5$ and fix a primitive fifth root of unity ω , so that $\omega^5 = 1$ and

$$1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0.$$

The minimal polynomial of ω is $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$, which is irreducible over $\mathbb{Q}$; hence

$$\mathbb{Z}[\omega] = \mathbb{Z}[x]/(x^4 + x^3 + x^2 + x + 1),$$

and $1, \omega, \omega^2, \omega^3$ form a $\mathbb{Z}$ -basis. Every quantity below is computed exactly in this ring: elements are integer 4-tuples (c_0, c_1, c_2, c_3) standing for $c_0 + c_1\omega + c_2\omega^2 + c_3\omega^3$, and the relation $\omega^4 = -(1 + \omega + \omega^2 + \omega^3)$ reduces products. No floating-point arithmetic is used anywhere, in this document or in the accompanying programs. In particular, “ $\widehat{f}(\xi) = 0$ ” always means exact vanishing in $\mathbb{Z}[\omega]$, and all equalities of transform values are ring equalities in $\mathbb{Z}[\omega]$.

3 The witness and its exact transform

Definition 2 (the witness). Let $f : \mathbb{F}_5 \rightarrow \mathbb{Z} \subseteq \mathbb{Z}[\omega]$ be

$$f = \delta_0 - \delta_1, \quad f(0) = 1, \quad f(1) = -1, \quad f(2) = f(3) = f(4) = 0.$$

Lemma 3 (support). $\# \text{supp}(f) = 2$, and $\text{supp}(f) = \{0, 1\}$.

Proof. Immediate: f is nonzero exactly at $x = 0$ and $x = 1$. □

Theorem 4 (exact transform). For every $\xi \in \mathbb{F}_5$,

$$\widehat{f}(\xi) = \sum_{x=0}^4 f(x) \omega^{\xi x} = \omega^0 - \omega^\xi = 1 - \omega^\xi.$$

Consequently $\widehat{f}(\xi) = 0$ if and only if $\xi = 0$, and $\# \text{supp}(\widehat{f}) = 4$.

Proof. Only $x = 0$ and $x = 1$ contribute to the sum:

$$\widehat{f}(\xi) = f(0)\omega^0 + f(1)\omega^\xi = 1 - \omega^\xi.$$

If $\xi = 0$ then $\widehat{f}(0) = 1 - 1 = 0$. Conversely suppose $\widehat{f}(\xi) = 0$, i.e. $\omega^\xi = 1$. Since $\omega^5 = 1$, write $\xi \in \{1, 2, 3, 4\}$; then ω is a root of both $x^5 - 1$ and $x^\xi - 1$, hence a root of $x^{\gcd(5, \xi)} - 1 = x - 1$ (as 5 is prime and $1 \leq \xi \leq 4$), so $\omega = 1$. But $\omega = 1$ is not a root of $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$, since $\Phi_5(1) = 5 \neq 0$; equivalently $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ would give $5 = 0$ in $\mathbb{Z}[\omega]$. Hence $\widehat{f}(\xi) \neq 0$ exactly for $\xi \in \{1, 2, 3, 4\}$, and $\# \text{supp}(\widehat{f}) = 4$. □

The transform values, written out in the exact basis of $\mathbb{Z}[\omega]$, are

ξ	$\widehat{f}(\xi)$
0	0
1	$1 - \omega = (1, -1, 0, 0)$
2	$1 - \omega^2 = (1, 0, -1, 0)$
3	$1 - \omega^3 = (1, 0, 0, -1)$
4	$1 - \omega^4 = 2 + \omega + \omega^2 + \omega^3 = (2, 1, 1, 1)$

using $\omega^4 = -(1 + \omega + \omega^2 + \omega^3)$. Exactly one of the five values is 0 and four are nonzero.

Corollary 5 (the witness is an equality configuration).

$$\# \text{supp}(f) + \# \text{supp}(\widehat{f}) = 2 + 4 = 6 = 5 + 1 = q + 1.$$

So f attains equality in (1), and its support pair is

$$(\# \text{supp}(f), \# \text{supp}(\widehat{f})) = (2, 4),$$

not the claimed $(q, 1) = (5, 1)$.

Remark 6. The value $\widehat{f}(4) = 1 - \omega^4$ shows that $\widehat{f}$ is *not* integer-valued: its ω -coefficient is 1. This is why the exact $\mathbb{Z}[\omega]$ model is needed and why no floating-point approximation can certify the support.

4 f is not a Fourier translate of an affine function

For $h : \mathbb{F}_5 \rightarrow \mathbb{Z}[\omega]$ and $\alpha, \beta \in \mathbb{F}_5$ define the *time–frequency translate*

$$(T_{\alpha, \beta} h)(x) = \omega^{\beta x} h(x + \alpha). \quad (2)$$

These are the “Fourier translates” of the conjecture: translation moves the argument and modulation multiplies by the character $\omega^{\beta x}$.

Lemma 7 (zero sets of affine functions). *For $a, b \in \mathbb{F}_5$ let $g_{a,b}(x) = ax + b$. Then*

$$\#\{x : g_{a,b}(x) = 0\} = \begin{cases} 5, & a = b = 0 \quad (\text{the zero function}), \\ 0, & a = 0, b \neq 0 \quad (a \text{ nonzero constant}), \\ 1, & a \neq 0 \quad (\text{exactly } x = -b a^{-1}). \end{cases}$$

Proof. If $a = 0$ the map is constant, equal to b ; it vanishes everywhere when $b = 0$ and nowhere otherwise. If $a \neq 0$ it is a bijection of $\mathbb{F}_5$, so it has exactly one zero, namely $x = -b a^{-1}$. $\square$

Lemma 8 (translates preserve the zero set up to a shift). *For any h and any α, β ,*

$$\{x : (T_{\alpha, \beta} h)(x) = 0\} = \{x : h(x) = 0\} - \alpha,$$

so the number of zeros is invariant under $T_{\alpha, \beta}$.

Proof. $\omega^{\beta x} \neq 0$ in $\mathbb{Z}[\omega]$ for every x (it is a unit), so $\omega^{\beta x} h(x + \alpha) = 0$ iff $h(x + \alpha) = 0$ iff $x + \alpha$ is a zero of h ; substituting $y = x + \alpha$ gives the claim. $\square$

Theorem 9 (non-membership). *The witness $f = (1, -1, 0, 0, 0)$ is not a time–frequency translate of any affine function $g_{a,b}$. Equivalently, there are no $a, b, \alpha, \beta \in \mathbb{F}_5$ with*

$$\forall x \in \mathbb{F}_5 : \quad \omega^{\beta x} g_{a,b}(x + \alpha) = f(x) \quad \text{in } \mathbb{Z}[\omega].$$

Proof. The zero set of the witness is $\{x : f(x) = 0\} = \{2, 3, 4\}$, of size 3. By Lemma 7, an affine function $g_{a,b}$ has 5, 0, or 1 zeros according as it is the zero function, a nonzero constant, or has $a \neq 0$. By Lemma 8 every time–frequency translate $T_{\alpha, \beta} g_{a,b}$ has the same number of zeros, i.e. one of 5, 0, 1; none of these is 3. Hence $T_{\alpha, \beta} g_{a,b} \neq f$.

The same conclusion follows from support sizes: a nonzero affine function has support 4 (when $a \neq 0$) or 5 (when $a = 0, b \neq 0$), the zero function has support 0, and $T_{\alpha,\beta}$ preserves the support size; but $\#\text{supp}(f) = 2$. An exhaustive machine check over all $5^4 = 625$ tuples (a, b, α, β) confirms this with exact $\mathbb{Z}[\omega]$ arithmetic (see `reproduce.py` and the Lean development). $\square$

Remark 10. If one reads “Fourier translate” as allowing the transform of a translate rather than a translate itself, the conclusion is unchanged: for $a \neq 0$ the transform of $T_{\alpha,\beta}g_{a,b}$ has non-integer ω -coefficients, and for $a = 0$ it is a nonzero multiple of δ_0 , so in neither case can it equal the integer-valued witness $(1, -1, 0, 0, 0)$. This is also checked exhaustively.

Remark 11 (why $\mathbb{F}_3$ is not enough). Over $\mathbb{F}_3$ the analogue of the witness would be $(1, -1, 0)$. That function is *not* a valid counterexample: as a function $\mathbb{F}_3 \rightarrow \mathbb{F}_3$ it equals $x \mapsto x + 1$ (since $-1 \equiv 2$ and $0 + 1, 1 + 1, 2 + 1 = 0$ give values $1, 2, 0$), i.e. it is a translate of an affine function. Over $\mathbb{F}_5$, by contrast, $(1, -1, 0, 0, 0)$ has three zeros while no affine function has three zeros. This is why the counterexample is taken over $\mathbb{F}_5$.

5 Enumeration of all equality configurations on $\mathbb{F}_5$

To quantify how badly the classification fails we enumerate every function $f : \mathbb{F}_5 \rightarrow \{0, \pm 1\}$, of which there are $3^5 = 243$. For each we compute $\hat{f}$ exactly in $\mathbb{Z}[\omega]$ and test the equality case $\#\text{supp}(f) + \#\text{supp}(\hat{f}) = 6$.

Theorem 12 (the equality configurations). *Among the 243 functions $\mathbb{F}_5 \rightarrow \{0, \pm 1\}$ there are exactly 32 equality configurations $\#\text{supp}(f) + \#\text{supp}(\hat{f}) = 6$, distributed by support pair as follows:*

$\#\text{supp}(f)$	description	count
1	the ten functions $\pm\delta_p, p \in \mathbb{F}_5$	10
2	the twenty functions $\pm(\delta_p - \delta_q), p \neq q$	20
5	the two nonzero constants ± 1	2
	total	32

Every one of the 20 functions of type $(2, 4)$ is an equality configuration and is not a time–frequency translate of an affine function. Moreover, over the 242 nonzero functions the bound (1) holds with equality attained, the minimum of $\#\text{supp}(f) + \#\text{supp}(\hat{f})$ being 6.

Proof. This is a finite computation over 243 cases with exact $\mathbb{Z}[\omega]$ arithmetic, carried out by `reproduce.py`. The script builds $\hat{f}$ from the definitions using the reduction $\omega^4 = -1 - \omega - \omega^2 - \omega^3$, counts supports, and verifies:

- exactly 32 functions attain the bound, with pairs $(1, 5)$: 10, $(2, 4)$: 20, $(5, 1)$: 2;
- the 20 functions of type $(2, 4)$ are exactly $\pm(\delta_p - \delta_q)$ for the ten unordered pairs $\{p, q\}$, two signs each;
- none of those 20 is a time–frequency translate of any affine function, and none is the transform of one.

The count $32 = 10 + 20 + 2$ coincides with the directly identifiable families: the $\pm\delta_p$ have transforms of support 5, and the nonzero constants have transforms of support 1 (a nonzero multiple of δ_0). The remaining 20 have no affine description. $\square$

Corollary 13 (the conjecture is false). *Conjecture 00000001075 fails on both counts:*

1. *The witness $f = (1, -1, 0, 0, 0)$ is an equality configuration (Corollary 5) that is not a Fourier translate of an affine function (Theorem 9); its support pair is $(2, 4)$, falsifying the parenthetical “support pairs of q and 1” as an exact description of the equality cases.*
2. *There are 20 further equality configurations of type $(2, 4)$ (Theorem 12) that are likewise not affine Fourier translates, falsifying “no other equality configurations exist”.*

6 Context: the premise is not a general theorem

The inequality (1) is not true for arbitrary q ; only its prime-field case is used above. Concretely, let $q = p^k$ with $k \geq 2$ and let $H \leq \mathbb{F}_q$ be the subgroup of size p^{k-1} (the kernel of the trace form $\mathbb{F}_q \rightarrow \mathbb{F}_p$); the additive group of $\mathbb{F}_q$ is a $\mathbb{F}_p$ -vector space of dimension k , and H is a hyperplane. Its Fourier transform is supported on the annihilator $H^\perp$, another hyperplane. With $f = \mathbf{1}_H$ one gets

$$\#\text{supp}(f) + \#\text{supp}(\widehat{f}) = p^{k-1} + p < p^k + 1 = q + 1 \quad (k \geq 2),$$

so (1) fails. For example, over $\mathbb{F}_4$ the subgroup $\{0, 1\}$ and its annihilator both have size 2, giving $2 + 2 = 4 < 5$; this is verified in `reproduce.py` with the explicit model $\mathbb{F}_4 = \mathbb{F}_2[t]/(t^2 + t + 1)$ and its additive characters in $\{\pm 1\}$. For prime q the bound does hold for nonzero f , as the enumeration at $q = 5$ confirms (minimum 6 over the 242 nonzero functions of shape $\{0, \pm 1\}$). Choosing the prime $q = 5$ keeps the counterexample inside the regime where the premise is true, so the refutation is not an artifact of a false hypothesis.

7 Reproducibility

The exact computations are carried out by `reproduce.py` (pure Python 3, standard library only, no `sympy`, no floating point):

```
$ python3 reproduce.py
...
PASS: the witness disproves conjecture 00000001075 as stated.
```

It prints the witness transform, the exhaustive non-membership checks over 625 tuples, and the 243-function enumeration with the 10/20/2 split, and exits non-zero if any check fails.

The LaTeX document is built with

```
$ tectonic main.tex
```

and the Lean development with

```
$ cd lean4 && lake build
$ cd lean4 && lake env lean Check.lean
```

8 Formalisation

The refutation is formalised in Lean 4 in `lean4/`, core Lean only (`import Std`, no `Mathlib`, no `sorry`), with $\mathbb{Z}[\omega] = \mathbb{Z}[x]/(x^4 + x^3 + x^2 + x + 1)$ represented as integer 4-tuples and multiplication given by the exact reduction $\omega^4 = -(1 + \omega + \omega^2 + \omega^3)$. All proofs are closed by `decide/rw`. The main formal statements are

```

theorem witness_supp : suppInt witness = 2
theorem witness_fhat_supp : suppQ (dft witness) = 4
theorem witness_equality : suppInt witness + suppQ (dft witness) = 6
theorem witness_fhat_values : dft witness 0 = 0 ∧ ...

theorem not_tfTrans_affine :
  ∀ (a b α β : Fin 5), ¬ (∀ x : Fin 5,
    tfTrans (affineVec a b) α β x = scalar (witness x))

theorem not_tfTrans_affine_dft : ...
theorem conjecture_00000001075_refuted : ...

```

Check.lean prints #print axioms for every theorem; the audit shows that each depends only on propext, with no sorryAx and no Lean.ofReduceBool. See lean4/README.md for the statement table.

Status against the submission rules

Rule 3 requires each submission to contain the LaTeX source code, a PDF document, and a Lean 4 project.

- **LaTeX source** — present (main.tex), a standalone article using only geometry, amsmath, amssymb, amsthm, array, parskip, and xeCJK, designed to compile with tectonic main.tex.
- **PDF document** — build/main.pdf is produced by the Tectonic build; the source is self-contained and uses no external figures or non-standard packages.
- **Lean 4 project** — present under lean4/, with lean-toolchain (leanprover/lean4:v4.33.1), lakefile.toml (library Main), Main.lean, Check.lean, and lean4/README.md. It formalises the witness’s support values, the equality $\# \text{supp}(f) + \# \text{supp}(\hat{f}) = 6$, and the non-membership of the witness among time–frequency translates of affine functions. It is core Lean only and contains no sorry.
- **Reproduction** — reproduce.py is pure standard-library Python 3 and verifies all numerical claims exactly.